

Trainer Debrief — Lab 18

Multiple Regression and Model Diagnostics (Black Belt Only)

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Build and validate a multiple regression model of your capstone Y (A3, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-18-multiple-regression.xlsx

Full model R-sq ~ 0.83, adjusted R-sq ~ 0.82. Significant: ClampPressure (b ~ +2.05 kPa/bar, p<0.001), DwellTime (b ~ +21.4 kPa/s, p<0.001), ElectrodeTipAge (b ~ -0.012 kPa/stroke, p<0.001). AmbientTemp and Humidity are NOT significant and have VIF > 10 — they are collinear (r ~ 0.75); drop one. The reduced 3-X model has essentially the same adjusted R-sq with a cleaner residual plot. Residuals are normal and show no pattern vs fits, so the model assumptions hold. $Y = f(\text{clamp, dwell, tip age})$ is the transfer function you optimise in the DOE.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your final model has all VIFs under 5, residual plots show no pattern, adjusted R-squared is reported, the model has been validated on held-out data, and each coefficient is stated in business terms.

Where this sits in the capstone

BLACK BELT ONLY — the Green Belt course teaches simple linear regression with one X. Real processes have several interacting Xs. Multiple regression models them together, but a model that fits is not yet a model that is valid, so the diagnostics matter as much as the coefficients. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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